DAASI: Arabic Text Simplification Corpus for Israeli Government Documents

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Abstract

Arabic-speaking citizens in Israel routinely encounter government website documents written in Modern Standard Arabic (MSA), a variety that differs substantially from the Levantine dialect they use daily. This diglossia gap limits access to public services and creates informational inequity, especially for elderly citizens unfamiliar with modern language tools. We address this problem through three contributions. We target simplification within Modern Standard Arabic (MSA) itself, motivated by the diglossia gap between MSA and the spoken varieties readers use daily; we do not perform translation or adaptation into a spoken dialect. First, we present **DAASI** (**D**iglossia-**A**ware **A**rabic **S**implification for **I**nclusion): a new parallel corpus of 2,350 sentence pairs drawn from Israeli National Insurance Institute documents, constructed via GPT-4o draft simplification followed by expert human validation. Second, we conduct a sociolinguistic survey of 50 Arabic-speaking adults in Israel confirming that government documents are harder to comprehend than school-level texts. Third, we benchmark two Arabic transformer models and GPT-4o across multiple fine-tuning strategies and conduct human evaluations to assess simplified text. Our results confirm that existing Arabic NLP benchmarks do not generalize to the Israeli government domain and that DAASI fills a critical resource gap for inclusive Arabic NLP in a diglossic context.

1 Introduction

Over 473 million people speak Arabic worldwide, yet the Arabic-speaking population in Israel faces a specific and underexplored challenge: formal government communication is conducted in Modern Standard Arabic (MSA), a variety that differs sharply from the Levantine dialects spoken daily by most Arab citizens (Ferguson, 1959; Saiegh-Haddad, 2003). This diglossia gap means that legal forms, healthcare instructions, and administrative notices are, for many readers, closer to a foreign language than to their mother tongue.

Our survey of 50 Arabic-speaking adults in Israel confirms the severity of this gap empirically: participants rated government documents significantly harder to understand than school-level texts across all measured dimensions – vocabulary complexity, sentence clarity, and overall comprehension.

Despite growing interest in Arabic NLP, existing text simplification resources focus on literary and news text in MSA (Nassiri et al., 2023; Alhafni et al., 2024) and do not cover the government domain relevant to Arabic-speaking citizens of Israel, including several health- and disability-related administrative categories (e.g., medical committee decisions, disability benefits) that are part of the government forms we collect, though the corpus is not a clinical or medical-text corpus in the narrower sense. This motivates our central contribution: DAASI (Diglossia-Aware Arabic Simplification for Inclusion), a new parallel corpus of 2,350 sentence pairs built from real Israeli official documents, all sourced from the National Insurance Institute (Bituach Leumi) website (see Section 4).

The stakes of this problem extend well beyond linguistic inconvenience. Arabic is one of Israel's two official languages, and government agencies are legally required to publish documents and websites in Arabic alongside Hebrew (Wikipedia contributors, 2024). Accessibility provisions further mandate that official content be made available in formats suitable for people with disabilities. In practice, however, formal compliance with these requirements does not guarantee communicative accessibility: official Arabic texts are typically written in bureaucratic Modern Standard Arabic (MSA), a register characterised by long nominal constructions, complex subordination, and specialised vocabulary that differs markedly from the Levantine dialect spoken at home by most Arabic-

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speaking citizens of Israel, especially the elderly ones (Ayari, 1996; Hudson, 2002). This gap is not unique to Arabic – plain-language movements in many countries have demonstrated that formally translated or legally accurate documents can still be practically inaccessible to their intended audiences (Ondov et al., 2022; Sikka and Mago, 2020). This is particularly evident in a diglossic setting where the written standard and the spoken language are structurally distant varieties (Ferguson, 1959; Saiegh-Haddad and Schiff, 2025). Automatic text simplification has emerged as a practical NLP response to this class of problem, with growing work on clinical and administrative domains in English and European languages (Espinosa-Zaragoza et al., 2023); yet Arabic remains severely under-resourced in this domain.

We make the following contributions:

- we introduce DAASI, a parallel corpus for simplification within Modern Standard Arabic (MSA), targeting government documents directed at Arabic-speaking citizens of Israel; the corpus is motivated by the diglossia gap but does not perform dialect adaptation (see Section 1 for this distinction);
- user survey quantifying perceived comprehension difficulty in formal Arabic texts, motivating the simplification task;
- benchmarks of transformer and generative models, quantifying the domain gap and measuring the gains from fine-tuning with automatic and human evaluation.
- we release `Tabsit.ai`, an open web platform for automatic Arabic text simplification using a hybrid pipeline, which additionally offers an optional experimental dialect-adaptation mode as a separate feature (Section B).

We emphasize that DAASI addresses *intralingual simplification of MSA* (lexical, syntactic, and structural simplification of formal MSA into more accessible MSA), not translation or adaptation into spoken Levantine Arabic. Diglossia motivates the task – it explains why MSA government text is difficult for readers whose everyday spoken variety differs substantially from MSA – but the corpus itself consists of MSA-to-simplified-MSA sentence pairs. We view dialect-level adaptation as a related but distinct direction, and discuss it as a limitation below.

2 Background and Related Work

Text simplification. Text simplification transforms a complex source text into a simpler form while preserving its meaning (Saggion and Hirst, 2017). It involves lexical simplification (replacing rare words with common synonyms), syntactic simplification (splitting long sentences), and structural reorganization (North et al., 2025; Carroll et al., 1998). Research in English has shown clear comprehension gains for learners, individuals with reading difficulties, and non-native speakers (Carroll et al., 1998; Espinosa-Zaragoza et al., 2023). Recent work has extended simplification to clinical and legal domains, where the gap between expert register and lay comprehension is particularly consequential (Ondov et al., 2022; Agrawal and Carpuat, 2024). Large language models have further broadened the toolkit: few-shot and instruction-tuned systems now achieve competitive simplification quality across multiple languages without domain-specific parallel data (Martin et al., 2022; Feng et al., 2023). Evaluation has likewise matured, moving beyond surface readability metrics toward human-centred assessments of meaning preservation and comprehension gain (Agrawal and Carpuat, 2024; Guidroz et al., 2025).

Arabic diglossia. Ferguson's (1959) foundational concept of diglossia describes communities in which a high (formal) variety and a low (spoken) variety coexist with distinct social functions. In the Arabic case, MSA serves formal writing while regional dialects serve everyday speech. In Israel, the dominant spoken variety is Levantine Arabic, and research has confirmed that even highly educated readers show reduced comprehension of MSA government texts compared to dialect-proximate texts (Saiegh-Haddad, 2003; Abu-Rabia, 2000). The phonological, morphological, and lexical distances between MSA and spoken Arabic are substantial enough that children effectively encounter a second language when they first enter formal literacy instruction (Saiegh-Haddad and Schiff, 2025; Abu-Rabia et al., 2022). This distance is not uniform across dialects: Levantine Arabic has been identified as structurally closer to MSA than many other spoken varieties, yet comprehension deficits in formal written registers persist even among proficient adult readers (Saiegh-Haddad and Schiff, 2025). The implications extend well beyond the classroom: bureaucratic and legal documents written in MSA nom-

inal constructions and complex subordination remain opaque to a significant share of the adult Arabic-speaking population, motivating register-aware NLP interventions (Hudson, 2002; Saiegh-Haddad, 2003).

Arabic text simplification. Nassiri et al. (2023) developed the first lexical simplification model for MSA based on word frequency analysis. Al-Thanyyan and Azmi (2023) proposed a hybrid machine-translation and transformer-based approach. The SAMER corpus (Alhafni et al., 2024) contains 28,507 complex-simple sentence pairs from news and educational texts and is currently the largest public Arabic simplification dataset, but it does not cover the government or medical domain. The BAREC shared task (Elmadani et al., 2025) has further advanced readability assessment for Arabic, providing fine-grained difficulty annotations across 19 levels that can inform simplification targets. BAREC is a readability-assessment resource (a classifier assigning a difficulty level to a given text) rather than a text-to-text simplification system, and is not directly comparable to the generative simplification models we benchmark in Section 5; we nonetheless see BAREC-style difficulty scoring as a promising direction for automatically screening or targeting simplification output in future work. Parallel efforts on Arabic readability, including lexicon-based tools (Jiang et al., 2020) and eye-tracking corpora (Baazeem et al., 2025), have supplied the field with resources for modeling difficulty, yet domain coverage remains skewed toward literary and journalistic text. No existing Arabic simplification resource addresses the governmental register.

Arabic NLP models. AraT5 (Nagoudi et al., 2021) is a sequence-to-sequence transformer pretrained on 70 GB of Arabic data and is the strongest open-source Arabic generative model for paraphrase and simplification. AraBERT (Antoun et al., 2020) is a discriminative encoder used for classification and lexical selection. mT5 is a massively multilingual text-to-text transformer that supports Arabic (Xue et al., 2021). GPT-4o serves as a strong few-shot commercial baseline. These models differ substantially in their exposure to dialectal Arabic: AraT5 and AraBERT are trained predominantly on MSA corpora, while mT5's pre-training includes some dialectal web text but without explicit dialect supervision (Xue et al., 2021). MARBERT and CAMeLBERT models (Abdul-Mageed et al., 2021; Inoue et al., 2021) represent more recent attempts to cover dialectal variation, though neither has been evaluated on simplification tasks.

3 Quantifying the Comprehension Gap

We recruited 50 adult Arabic-speaking participants residing in Israel through a purposive sample stratified across three age groups and genders (see Table 1), three education levels (primary, high school, and higher), and diverse occupational backgrounds. All participants volunteered for the task, and their identifying information was protected by providing them anonymous IDs. All participants are native speakers of Levantine Arabic and are proficient in MSA. As Table 1 shows, our convenience sample is heavily skewed toward the 18–25 age group (36 of 50 participants); older adults, who arguably face the largest practical barriers to accessing government services, are underrepresented (only 6 participants aged 41+). We flag this imbalance explicitly as a limitation on the generalizability of our comprehension-gap findings (see Limitations) and identify a larger, age-stratified follow-up study as important future work.

Category	Count	Category	Count
Age 15–18	4	Female	34
Age 18–25	36	Male	16
Age 26–40	4		
Age 41+	6		

Table 1: Participant demographics.

We selected 100 texts, 50 of which are authentic government document passages and 50 of which are school-level descriptive passages. The school-level texts were authentic educational materials collected from the Ministry of Education website, corresponding approximately to middle school level. They were selected as a lower-complexity comparison group representing texts accessible to school-age readers, rather than as simplification targets. Each participant read one text from each group in a random order. After each text, they completed a Likert questionnaire (Nemoto and Beglar, 2014) on four parameters: (1) overall difficulty, (2) vocabulary complexity, (3) sentence clarity, and (4) overall comprehension. In this type of questionnaire, response options were presented as ordered text labels (e.g., "very easy" to "very difficult") rather than bare numbers; the number of response

options varied slightly by item (3–5 options depending on the question), and we code each item on its own ordinal scale from 1 (easiest / clearest / most comprehensible) to N (hardest / least clear / least comprehensible), where N is that item's number of response options. Direction is applied consistently across all four parameters, so that for every dimension a *higher* score indicates a *harder-to-process, less clear, or less comprehensible* text, never the reverse. Additionally, we included two further items: a closed checkbox question listing candidate reasons for comprehension difficulties (participants selected all that applied), and a separate free-text field for any additional comments. We refer to these consistently as the *closed-item* responses and *open-text* responses, respectively, in what follows.

Table 2 summarizes the mean Likert scores for all survey participants. We see that the government documents consistently received higher difficulty ratings across all four metrics. We tested each of the four government-vs-school comparisons using a Wilcoxon signed-rank test (participants rated one text from each group, so ratings are paired); all four differences are statistically significant at $p < 0.01$ (three of the four at $p < 10^{-6}$) (full scores appear in the Appendix). Figure 1 shows distribution of answers to the reasons for comprehension difficulties question.

Dimension	Gov. doc.	School text
Overall difficulty	3.64 ± 1.01	2.20 ± 1.07
Vocabulary complexity	2.86 ± 0.73	1.66 ± 0.80
Sentence clarity	2.88 ± 0.90	2.38 ± 1.05
Overall comprehension	2.22 ± 0.58	1.38 ± 0.64

Table 2: Mean $\pm$ SD perceived difficulty of survey documents (higher = more difficult / less clear / less comprehensible for all four rows; each row uses that item's own response-option range, from 1 to N, where N is 5 for Overall difficulty, 4 for Vocabulary complexity, 4–5 for Sentence clarity, and 3 for Overall comprehension).

We adopt mean($\pm$SD) as our Likert-reporting convention for this survey. Table 9's (Section 5.5) separate human evaluation survey reports only aggregated %$\geq$4 proportions, since its underlying raw per-participant ratings were not retained; we note this difference explicitly where Table 9 is presented.

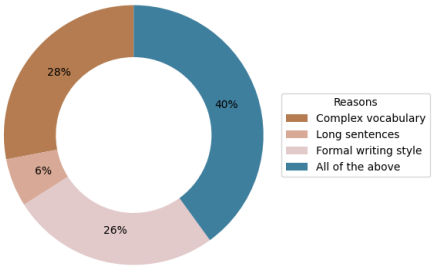


Figure 1: Distribution of factors leading to comprehension difficulty (according to survey participants).

Qualitative analysis of the open-text responses identified three recurring difficulty sources: (1) bureaucratic MSA vocabulary absent from everyday speech (e.g., استيفاء المتطلبات vs. the colloquial تحقيق الشروط), (2) long nominal chains typical of formal Arabic legal style, and (3) passive constructions with dropped agents that obscure responsibility. These findings directly motivate our text simplification strategy in Section B.

4 The DAASI Corpus

In this section, we describe the DAASI parallel simplification corpus which is the first such resource for Israeli government Arabic texts.

Data collection. We collected 50 official documents containing Arabic-language content directed at Arabic-speaking citizens of Israel from the official government website. We used the Arabic section of the National Insurance Institute (Bituach Leumi) as our primary source: https://www.btl.gov.il/Arabic%20HomePage/about_ar/forma_and_authorization_ar/Forms_ar/Pages/default.aspx.

Document categories
General; Unemployment; Domestic Violence; Maternity; Insurance Contributions; Work Injury Benefit; Accident Benefit; Income Support; Senior Citizen (Old Age); Children; Child Support (Alimony); Reserve Military Service; Rehabilitation; Burial Grant; Prisoners of Zion; Survivors (Widow/Widower and Orphans); Work Accident Victims; Hostile Action Victims; Sickness Benefit; Bankruptcy; Work Grant (Negative Income Tax / Employment Grant); Employees; Mobility Benefits; General Disability; Research Funding Assistance; Savings Program for Every Child; Special Services; Counseling Service for Senior Citizens and Their Families; Funds and Trusts; Medical Committees; Children with Cerebral Palsy; Volunteers; International Conventions; Disabled Child

Table 3: Document categories at the National Insurance Institute website.

This site contains more than 30 categories of forms and public-service documents (see Table 3). We selected documents from these categories and extracted the Arabic content available on the official website. The collected documents include forms, eligibility information, public service instructions, and citizen guidance materials intended for Arabic-speaking users. Documents were manually segmented into individual sentences, yielding 2,702 source sentences. Segmentation was performed by the two human annotators described below, following a written guideline instructing them to (i) split at sentence-final punctuation (periods, question marks) and at semicolons introducing an independent clause, (ii) keep list items introduced by a shared stem together with that stem as a single unit rather than splitting mid-list, and (iii) exclude non-sentential content such as form field labels, headers, and tables of numeric values, which were removed prior to segmentation.

Simplification and annotation. Corpus construction was performed in four stages:

1. **Automatic draft.** First, we automatically simplified each sentence using GPT-4o using a few-shot prompt instructing the model to (a) replace formal MSA vocabulary with everyday Arabic equivalents, (b) split sentences longer than 20 words, and (c) retain the original meaning (see Section 8).

2. **Human validation.** Two native Arabic-speaking annotators – both proficient in MSA and Levantine Arabic – reviewed every simplified sentence for accuracy, naturalness, and meaning preservation. Disagreements were resolved by a third independent annotator.

3. **Alignment and filtering.** Sentence pairs where the simplification was judged to change the meaning or omit key information were discarded, yielding 2,350 final pairs.

To make the extent of human correction to the GPT-4o draft transparent: of the 2,702 candidate simplifications, 352 (2,702 − 2,350) were discarded during alignment and filtering because the simplification was judged to change the meaning or omit key information (Section 4); the remaining 2,350 pairs were retained after human review, with corrections ranging from none (accepted as-is) to substantive revision. We did not retain a fine-grained accept/revise/discard breakdown at the level of individual edits, which we note as a limitation; future iterations of this corpus will log this breakdown explicitly. Disagreements between the two primary annotators were referred to a third independent annotator, whose decision was final.

Corpus analysis. To characterize the corpus, we computed descriptive statistics for both the original and simplified sentences. Average character and word length was calculated directly from the text, while average subword token count was measured using the tokenizer of AraBERT v0.2 (Antoun et al., 2020). We also report total word, unique word, and unique lemma counts computed using the Arabic pipeline of the Stanza NLP toolkit (Qi et al., 2020). The results are shown in Table 4.

Measure	Original	Simplified
avg char. length	41.14	41.34
avg word count	10.95	10.70
avg subword count	15.68	15.09
total unique words	3245	2890
total unique lemmas	1918	1622

Table 4: DAASI corpus statistics.

We evaluate lexical diversity in original and simplified texts of the DAASI corpus using four standard metrics: Type–Token Ratio (TTR) (Carter and McCarthy, 2014), which measures the proportion of unique word types; HD-D, a robust estimate of lexical diversity based on probabilistic sampling; MTLD (Measure of Textual Lexical Diversity) (McCarthy and Jarvis, 2010), which captures vocabulary variation independent of text length; and Yule's K, a measure of lexical repetition where lower values indicate higher diversity (Williams, 1946). The results in Table 5 show that the simplified corpus exhibits a slightly lower TTR alongside a lower (i.e., more diverse, since lower Yule's K indicates less repetition) Yule's K, and marginal increases in HD-D and MTLD. Taken together we interpret this as lexical diversity being essentially preserved after simplification: the length-independent metrics (HD-D, MTLD, Yule's K) all move in the direction of equal-or-greater diversity, while the small TTR decrease is consistent with TTR's known sensitivity to text length rather than an actual drop in diversity. Readability, measured using an Arabic-adapted Flesch-like metric (where higher scores indicate easier text), improves in the simplified texts, which obtain higher scores and are

Lexical Diversity & Readability			Lexical and Syntactic Complexity		
Metric	**Original**	**Simplified**	**Metric**	**Original**	**Simplified**
TTR	0.0881	0.0815	Noun Ratio	0.4254	0.3769
HD-D	0.6785	0.6830	Verb Ratio	0.0511	0.0795
MTLD	11.7198	11.9494	Adjective Ratio	0.0945	0.0746
Yule's K	373.5122	371.0642	Dependency Depth	5.1203	4.9305
Flesch Score	-209.8095	-190.3803	Content Word Ratio	0.5710	0.5310

Table 5: Lexical and syntactic complexity and readability of the DAASI corpus.

therefore easier to process. This metric adapts the classic Flesch Reading Ease formula to Arabic by substituting characters per word for syllables per word, following (El-Haj and Rayson, 2016).

To assess lexical and syntactic complexity, we compute the proportions of nouns, verbs, and adjectives, the content word ratio, and dependency depth using the Stanza NLP toolkit (Qi et al., 2020). As shown in Table 5, simplified texts exhibit reduced linguistic complexity, with lower noun and adjective ratios, higher verb ratio, lower content word ratio, and reduced dependency depth, indicating simpler and less syntactically embedded structures.

To evaluate semantic consistency within and across text versions, we compute embedding-based cosine similarity scores for original and simplified texts as well as between them. The results in Table 6 show high semantic consistency between the two versions in the DAASI dataset, with inter-group similarity comparable to intra-group scores, indicating that the simplification process largely preserves meaning. The slightly higher intra-group similarity in the simplified corpus suggests reduced linguistic variation after simplification.

Metric	**Avg cosine similarity**
original intra-group	0.4884
simplified intra-group	0.5068
original vs simplified	0.4917

Table 6: Semantic similarity analysis of the DAASI dataset.

5 Experimental Evaluation

Figure 2 illustrates the evaluation setup used in this study.

We split the DAASI dataset with 2,350 sentences as 80/10/10 train/validation/test split and fine-tune transformer models on the DAASI train set, optionally enhanced with the SAMER dataset (Alhafni et al., 2024). The SAMER dataset (Alhafni et al., 2024) is a large-scale Arabic text simplification corpus containing pairs of original and simplified texts, which we use both as an external source of simplification knowledge and to examine the impact of domain differences between SAMER and DAASI on model performance and generalization. The SAMER dataset is used in full or in a form of a random sample of the exact same size as the DAASI train set to ensure balanced representation of both types of data.

For the non-trainable LLM with prompting (GPT-4o), we use zero-shot and few-shot prompts. The zero-shot prompt is nearly identical to the base prompt (see Section 8) except for examples; for the few-shot setup, it is enhanced by examples from the DAASI training set.

5.1 Models

We evaluate four models:

1. AraT5 (Nagoudi et al., 2021) (`UBC-NLP/AraT5-base` on HuggingFace). A sequence-to-sequence transformer specifically pre-trained for Arabic text generation and understanding tasks, following the T5 text-to-text paradigm.

2. mT5 (Xue et al., 2021) (`google/mt5-base` on HuggingFace). A massively multilingual text-to-text transformer pre-trained on 101 languages and designed for a wide range of multilingual generation and understanding tasks.

3. GPT-4o (Hurst et al., 2024) in a zero-shot setting, denoted by GPT ZS. Prompted directly to simplify text without task-specific examples, relying solely on the task instructions.

4. GPT-4o (Hurst et al., 2024) in a few-shot setting, denoted by GPT FS. Prompted with input-output simplification examples from the DAASI dataset.

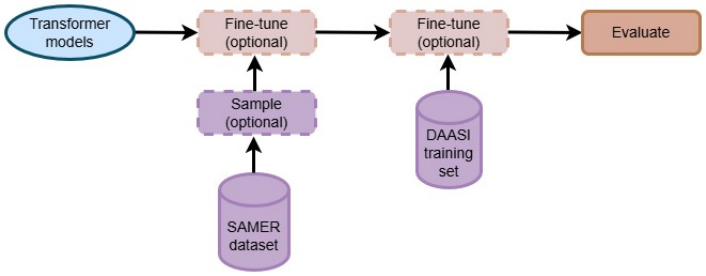


Figure 2: Evaluation pipeline.

Transformer models (AraT5 and mT5) are fine-tuned on training data for 10 epochs, batch size 16, with learning rate 5×10^{-5}.

5.2 Metrics

We report *BLEU* (Papineni et al., 2002), which measures n-gram overlap with human references, and *SARI* (Xu et al., 2016), which rewards appropriate additions, deletions, and paraphrases and is the standard evaluation metric for text simplification. We additionally report *BERTScore* (Zhang et al., 2020) (F1), computed with AraBERT v0.2 embeddings (Antoun et al., 2020) (9 layers, no baseline rescaling), as a meaning-preservation metric that is not tied to surface n-gram overlap and therefore complements BLEU/SARI, particularly for systems such as GPT-4o that rewrite aggressively. We also report *%unchanged*, the ratio of output sentences identical to the source, as a measure of simplification aggressiveness.

5.3 Automatic Evaluation Results

Table 7 presents the performance of AraT5 and mT5 models under different fine-tuning strategies, evaluated on the DAASI test set.

The results show that fine-tuning on the simplification-oriented DAASI dataset consistently yields better performance than training on the SAMER datasets alone. For both transformer models, the best results are achieved by first fine-tuning on the SAMER and DAASI datasets, with AraT5 obtaining the highest scores while also producing the lowest percentage of unchanged outputs (15.74%). In contrast, models trained only on SAMER exhibit substantially higher unchanged rates, indicating a tendency to preserve the original text rather than perform effective simplification, which highlights the domain gap between the two datasets and the importance of task-specific adaptation through subsequent fine-tuning on DAASI.

The 0% unchanged rate for GPT-4o means it rewrites every sentence, which reduces BLEU against a conservative gold reference.

5.4 Linguistic Analysis

We report the performance of the best-performing system (AraT5 fine-tuned on the combined SAMER and DAASI datasets), across lexical, readability, syntactic, and content-related metrics in Table 8. Overall, the results indicate that the model consistently preserves lexical diversity (i.e., diversity is not reduced relative to the gold standard) while producing texts that are easier to read and syntactically simpler than the original inputs. These trends align with effective simplification, showing reduced structural complexity and controlled changes in lexical usage without loss of semantic content.

Metric	Original	Gold Standard	Predicted
TTR	0.2079	0.1944	0.1918
HD-D	0.7363	0.7379	0.7251
MTLD	17.9541	18.1443	17.0296
Yule's K	277.5530	278.2512	305.2959
Flesch-like	-203.7940	-183.2127	-194.9636
noun ratio	0.4160	0.3707	0.3837
verb ratio	0.0520	0.0840	0.0709
adjective ratio	0.0983	0.0708	0.0900
dep. depth	5.2447	5.0106	4.9894

Table 8: Lexical diversity and complexity metric comparison on the DAASI test set.

5.5 Human Evaluation

The same 50 survey participants from Section 3 evaluated matched original and automatically simplified sentence pairs from the DAASI test set (we used the output of our best model). Participants rated each sentence on three 5-point Likert scales: comprehension, fluency, and vocabulary simplicity. They also indicated their preference between the original and simplified versions.

Model	FT on	BLEU	SARI	BERTScore	% unchanged
AraT5	DAASI	41.83	59.42	81.19	17.02%
AraT5	SAMER (sample)	30.20	45.03	76.45	56.81%
AraT5	SAMER	29.93	46.79	77.39	66.81%
AraT5	SAMER+DAASI	44.73	63.20	83.76	15.74%
AraT5	SAMER (sample)+DAASI	43.71	61.50	83.05	16.81%
mT5	DAASI	37.29	53.70	77.16	24.04%
mT5	SAMER (sample)	30.65	44.04	76.43	56.81%
mT5	SAMER	33.43	49.38	77.26	85.32%
mT5	SAMER+DAASI	39.97	56.55	79.97	31.49%
mT5	SAMER (sample)+DAASI	38.15	53.78	79.36	29.57%
GPT-4o ZS	–	25.64	52.25	78.26	0.00%
GPT-4o FS	–	29.00	54.70	80.00	0.02%

Table 7: Evaluation results (best scores are marked in gray).

As can be seen from Table 9, the automatically simplified sentences received substantially higher ratings across all three dimensions. We indicate there %≥4 – percentage of ratings in the top two points on the 5-point scale. As a result, 84% of ratings for simplified sentences fell in the 4-5 range, compared to 35% for originals on the comprehension dimension. The strongest effect was for vocabulary simplicity: ratings in the 4-5 range rose from 31% (original) to 88% (simplified), directly confirming the advantages of our best model.

Metric	Original (%≥4)	Simplified (%≥4)
Comprehension	35	84
Fluency	42	81
Vocabulary simplicity	31	88
Preference: simplified	84%	

Table 9: Human evaluation results (1–5 scale; higher = better for all three rows here, the opposite convention from Table 2, since this table measures positive quality rather than difficulty – noted explicitly to avoid the ambiguity raised in review)

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These are the exact pooled %≥4 values reported in our final human-evaluation summary. We report only this statistic for Table 9 rather than a mean(+SD), as we did for Table 2: the underlying raw per-participant ratings for this particular evaluation were not retained, only the aggregated %≥4 figures, which we note as a limitation on the granularity of this specific analysis.

6 Discussion and Conclusion

Our results show that the main challenge in Arabic government text simplification is the domain mismatch between existing resources and the target register. Models trained only on SAMER perform poorly on DAASI, with AraT5 achieving BLEU 29.93, SARI 46.79, and a high unchanged rate (66.81%), indicating limited simplification behavior. In contrast, fine-tuning on DAASI improves performance substantially (BLEU 44.73, SARI 63.20, 15.74% unchanged), demonstrating the importance of domain-specific data. The gap reflects the formal MSA register, bureaucratic vocabulary, and institution-specific terminology present in government documents.

GPT-4o provides competitive zero-shot and few-shot simplification performance (SARI 52.25 and 54.70, respectively), but the best AraT5 configuration outperforms it while offering a locally deployable and auditable alternative. Human evaluation further confirms that DAASI-based simplification improves comprehension and vocabulary accessibility.

In this work, we introduced DAASI, a government-domain Arabic simplification corpus motivated by the comprehension gap between formal MSA documents and the needs of Arabic-speaking citizens in Israel. Our experiments demonstrate that targeted training data is critical for effective simplification, and we release DAASI and the Tabsit.ai platform to support future research on inclusive Arabic NLP.

7 Limitations

DAASI covers only 50 documents from Israeli government authority, which is modest compared to SAMER's 28,507 pairs, and is therefore limited in domain breadth. The current size reflects the cost of expert human validation, which we treat as necessary for a resource intended for public-facing accessibility use; scaling further will require either additional annotator time or a lighter-weight quality-filtering model trained on the existing validated pairs to pre-screen future GPT-4o drafts before human review, which we leave to future work. Human validation was performed by two annotators; Inter-annotator agreement was not recorded during corpus construction, so we cannot retrospectively report an agreement statistic for the Accept/Revise/Reject decisions. The platform currently serves Levantine Arabic speakers in Israel and has not been evaluated for other dialects. Our comprehension survey (Section 3) also relies on a convenience sample skewed toward younger adults (Table 1); a larger, age- and education-stratified sample would strengthen the generalizability of the reported comprehension gap.

The Tabsit.ai system was evaluated only with Levantine speakers; generalization to Egyptian, Gulf, or Moroccan dialect speakers has not been tested.

Finally, only aggregated summary statistics were retained for the original-vs-simplified human evaluation (Table 9); the raw per-participant ratings were not preserved, so we cannot report means, standard deviations, or a paired significance test for that particular survey, unlike the government-vs-school comprehension survey (Table 2) for which raw ratings were available.

8 Data and Model Availability

The data is publicly available at https://github.com/NataliaVanetik/DAASI-simplification-corpus, and the trained model and application are available at https://github.com/Huseen1141/tabsit. This GitHub project also contains full annotation guidelines and the base simplification prompt used in this work.

9 Ethical Considerations

Survey participants provided written informed consent and were made aware of the research purpose. No personally identifiable information was collected or retained. Participants volunteered for the task. Corpus documents were collected from publicly available Israeli government websites and used solely for academic research. Annotation was performed by native Arabic speakers who volunteered for the task. The Tabsit.ai application is designed to assist Arabic-speaking citizens in accessing public information; it does not alter legal documents or produce authoritative translations.

We use "Arabic-speaking citizens of Israel" to describe the population whose comprehension of government communication we study; this is a citizenship-based description and is not intended to imply a single linguistic, religious, or political identity – it may include, for example, Mizrahi and Sephardi Jewish citizens who also speak Arabic dialects, alongside Palestinian citizens of Israel. We use "Levantine Arabic" to refer to the spoken variety our participants speak, while noting that this label spans a broader region than Israel alone and that "Palestinian Arabic" is the more precise term for many of our participants specifically. We avoid the term "Arab Israeli" in the body of this paper, since it is rejected by many Palestinian citizens of Israel as eliding Palestinian national identity; we retain the term only where it appears verbatim in cited source material. We acknowledge that these three dimensions – citizenship, linguistic variety, and identity – are analytically distinct and should not be treated as interchangeable.

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Appendix

A Likert Score Standard Deviations and Significance Tests

Item coding. Each survey item's response options were coded on its own ordinal scale from 1 (easiest/clearest/most comprehensible) to N (hardest/least clear/least comprehensible), where N is that item's number of distinct response options: Overall difficulty ($N = 5$), Vocabulary complexity ($N = 4$), Sentence clarity ($N = 4$ for the government-text item, $N = 5$ for the school-text item, which offered an additional "very clear" option), and Overall comprehension ($N = 3$).

Table 2 means with standard deviations:

Dimension	Gov. (mean±SD)	School (mean±SD)
Overall difficulty	3.64 ± 1.01	2.20 ± 1.07
Vocabulary complexity	2.86 ± 0.73	1.66 ± 0.80
Sentence clarity	2.88 ± 0.90	2.38 ± 1.05
Overall comprehension	2.22 ± 0.58	1.38 ± 0.64

Wilcoxon signed-rank tests (government vs. school, paired by participant, $n = 50$):

Dimension	Statistic	p-value
Overall difficulty	$W = 53.5$	$p = 4.5 \times 10^{-7}$
Vocabulary complexity	$W = 28.5$	$p = 1.3 \times 10^{-7}$
Sentence clarity	$W = 159.5$	$p = 9.1 \times 10^{-3}$
Overall comprehension	$W = 12.5$	$p = 3.3 \times 10^{-7}$

All four comparisons are statistically significant ($p < 0.01$); three of the four are significant well beyond conventional thresholds ($p < 10^{-6}$). The sentence-clarity comparison, while still significant, shows a smaller effect than the other three dimensions.

Table 9's (Section 5.5) original-vs-simplified human evaluation is a separate survey instrument from the government-vs-school comprehension survey analyzed above. Only the aggregated %≥4 values for that evaluation were retained; the raw per-participant ratings needed to compute means, standard deviations, and a paired significance test for that survey do not exist, which we note as a limitation.

B Tabsit.ai application

Tabsit.ai is a full-stack web application for Arabic text simplification. It follows a three-tier architecture consisting of a React/Node.js frontend, a Python FastAPI backend, and a simplification engine combining AraT5 (Nagoudi et al., 2021) and GPT-4o. Users can paste or upload Arabic text and receive side-by-side original and simplified versions. Additional features include an educational mode that highlights replaced terms with explanations and a dialect mode that adapts output toward Levantine Arabic. The simplification engine employs a two-stage hybrid pipeline. First, syntactic simplification uses Stanza's Arabic parser to identify sentence boundaries and part-of-speech tags, splitting overly long sentences at coordinating conjunctions (و, ف, لكن) and embedded clauses. Second, lexical simplification replaces difficult vocabulary using a lexicon of over 2,000 formal-colloquial word pairs derived from survey responses and administrative glossaries, with substitutions validated through mBERT pseudo-log-likelihood scoring (Devlin et al., 2019). For passages classified as highly complex by an AraT5-based difficulty classifier, GPT-4o few-shot generation serves as a fallback to simplify constructions not covered by the rule-based pipeline.

C Segmentation and Annotation Guidelines

Sentence segmentation. Documents were segmented by the two annotators using the following rules: (i) split at sentence-final punctuation (periods and question marks) and at semicolons introducing an independent clause; (ii) retain list items introduced by a shared stem with that stem as a single unit when the items are not independently comprehensible; (iii) exclude non-sentential content, such as form labels, headers, and tables of numeric values; and (iv) do not treat abbreviations or numbered references (e.g., article or paragraph numbers) as sentence boundaries.

Accept / revise / reject guideline for GPT-4o drafts. For each candidate simplification from the automatic-draft stage (see the prompt document), annotators classified it as **Accept** if it preserved all factual content (names, dates, numbers, and deadlines), was grammatically correct and natural, and was easier for a non-specialist reader to understand. A draft was marked **Revise** if it preserved the meaning and was directionally simpler but required only a minor correction, such as fixing an unnatural word choice, restoring an omitted number, or replacing an overly literal bureaucratic expression. A draft was **Reject** if it changed the meaning, omitted required factual content, or failed to simplify the sentence; such pairs were either regenerated using a revised prompt or discarded. Disagreements were resolved by a third independent annotator, whose decision was final.